Experiment-8

Program to Implement DFS

Aim

To develop a Python program to implement the Depth-First Search (DFS) algorithm for graph traversal.

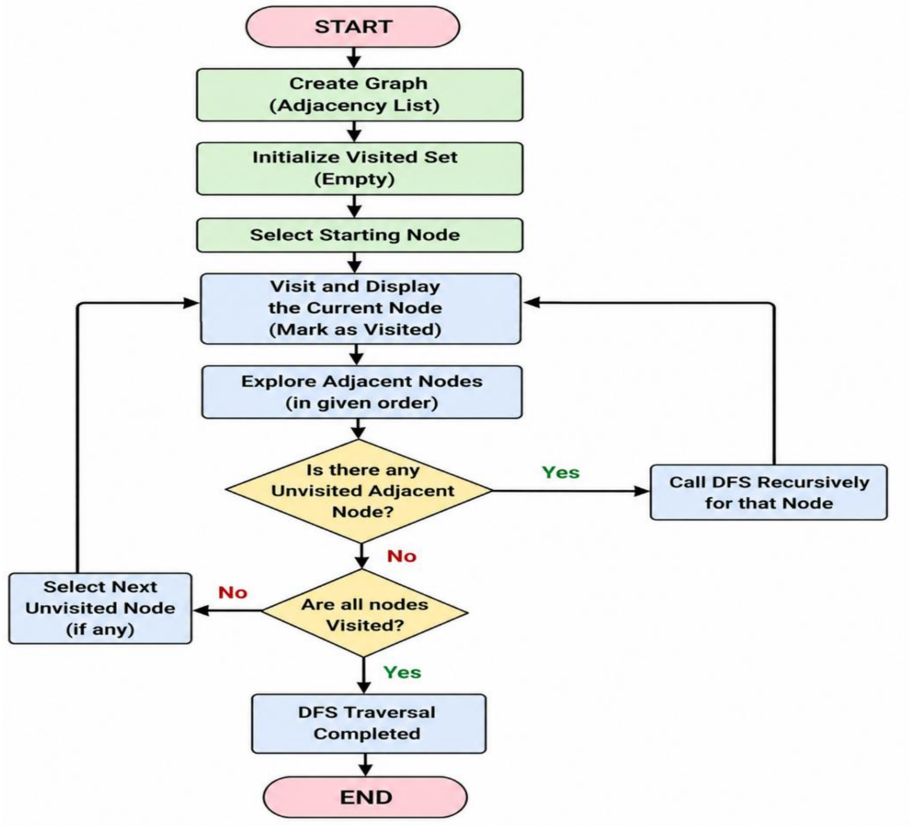
Objective

- To implement the Depth-First Search (DFS) algorithm.
- To traverse all nodes of a graph using depth-first traversal.
- To understand graph traversal techniques in Artificial Intelligence.

Algorithm

1. Start the program.
2. Create the graph using an adjacency list.
3. Initialize an empty visited set.
4. Start DFS from the source node.
5. Mark the current node as visited and display it.
6. Recursively visit all unvisited adjacent nodes.
7. Repeat until all reachable nodes are visited.
8. Stop the program.

Flowchart:

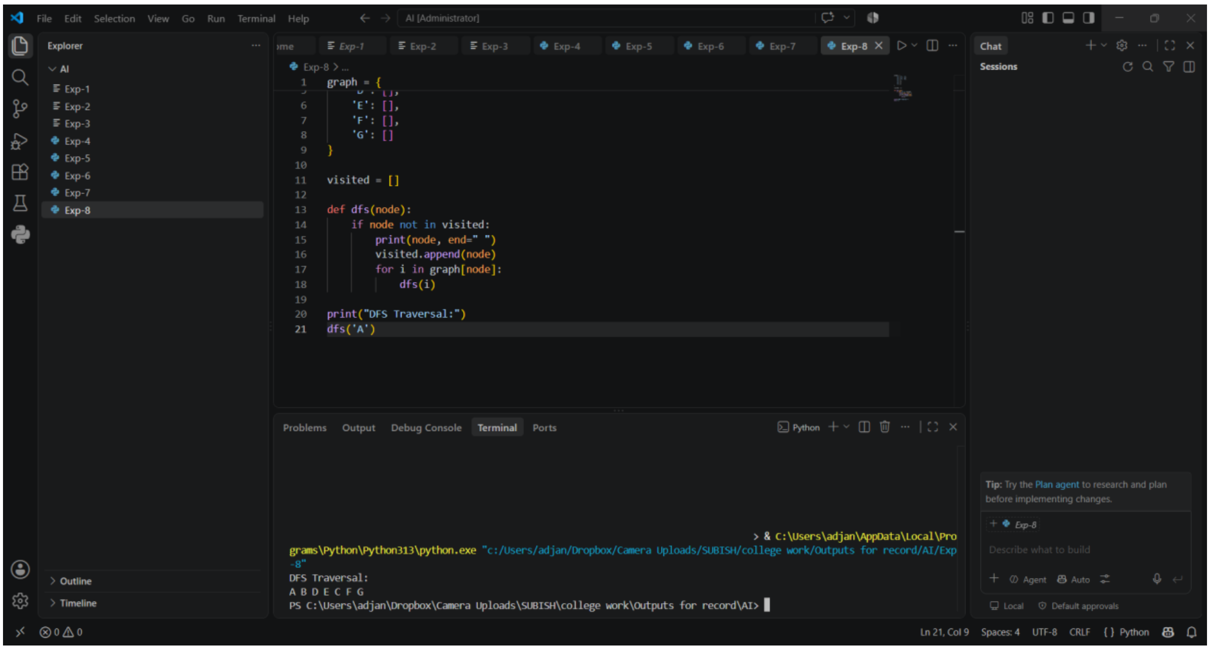


Code:

```
graph = {
    'A':['B','C'],
    'B':['D','E'],
    'C':['F','G'],
    'D':[], 'E':[], 'F':[], 'G':[]
}

visited = set()

def dfs(node):
    if node not in visited:
        print(node, end=" ")
        visited.add(node)
        for i in graph[node]:
```

```
            dfs(i)

dfs('A')
```

Output:



Result:

The Depth-First Search (DFS) algorithm was successfully implemented, and all graph nodes were traversed in depth-first order.

Conclusion:

The Depth-First Search (DFS) algorithm efficiently traverses a graph by exploring each branch completely before backtracking, making it suitable for graph search and traversal problems.